

NeuFlow v3: Queryable Optical Flow with Calibrated Uncertainty for Region-Directed Perception

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Abstract—Optical flow networks emit a dense displacement field at a single fixed resolution. Many consumers of flow, including image registration, sparse tracking and survey mosaicking, require a few hundred correspondences at locations they select rather than a value at every pixel. We present NeuFlow v3, which retains the NeuFlow v2 pipeline through its coarse flow stage and replaces only the final convex upsampler with an implicit decoder evaluated at arbitrary continuous coordinates. We pursue three objectives: queryable output, so cost scales with the number of queries, $O(N)$, rather than with image area, $O(H \times W)$; cheap repeatability, so a frame pair is processed once and every subsequent query is answered from cached state at 1.25 ms against 33.3 ms, a factor of 27; and calibrated confidence, a per-query error scale the baseline cannot express at all. The decoder is 745,226 parameters, 38% of the stage it replaces, and the complete model is 13.3% smaller than the baseline. On held-out driving sequences it is 2.6% less accurate on mean end-point error. The confidence signal is the more consequential capability: ranking queries by predicted error lets the model abstain on its least confident 20%, reducing mean error over the remaining 80% of the frame to 1.06 px, a selection the baseline cannot make at any threshold since it emits flow alone. We further characterise region-directed inference for fast-moving platforms, showing that the context margin a cropped region requires is set by inter-frame displacement rather than image size, a relation we verify at two motion scales differing by a factor of three.

Index Terms—optical flow, implicit neural representation, uncertainty estimation, edge computing, region of interest

I. Introduction

Optical flow estimation has converged on a common output contract: given two frames, produce a displacement vector for every pixel, at the resolution of the input. Modern real-time networks such as NeuFlow v2 [1] attain this efficiently enough for embedded deployment. The contract itself, however, is rarely questioned, and it does not match what several important consumers of flow actually require.

Image registration selects a few hundred well-textured points and needs correspondences at those points. Sparse feature tracking needs flow at detected keypoints. Survey mosaicking needs matches within overlap regions, frequently at sub-pixel positions. In each case a dense field is computed, and the great majority of it is discarded. On constrained hardware, computing 479,232 correspondences

in order to consume 800 of them is the difference between operating in real time and not.

The fixed-resolution contract also forecloses two operations outright. A consumer cannot request flow at a position between input pixels, and it cannot request additional correspondences on a frame the network has already processed without paying for the entire computation again.

This paper asks whether the final stage of a real-time flow network can be made queryable without surrendering accuracy or throughput, and reports what such a change costs and what it buys. We pursue three objectives, each of which the results section addresses in turn:

- 1) **Queryable output.** An implicit decoder replaces NeuFlow v2’s convex upsampler and answers flow queries at arbitrary continuous coordinates in $O(N)$ rather than $O(H \times W)$, with sparse output identical to dense output at matching coordinates to 0.00057 px.
- 2) **Cheap repeatability.** The network is factored into a coarse pass computed once per frame pair and cached, and a decode evaluated against that cached state. A repeat query costs 1.25 ms against the baseline’s 33.3 ms full recomputation, a factor of 27, because the baseline retains nothing between calls.
- 3) **Calibrated confidence.** A per-query error scale b under a Laplace likelihood, monotonic over a $15\times$ range of predicted values, which enables selective prediction: $2.2\times$ lower error than the baseline over 80% of the frame. The baseline emits flow alone and cannot abstain at any price.

Two further results accompany them. We characterise region-directed inference for fast platforms, including a design rule relating the required context margin to platform speed and frame rate, verified across two domains whose motion scales differ by a factor of three. And we account explicitly for what the change costs: 2.6% of mean accuracy on the training domain, with the mechanism identified.

II. Related Work

Variational and local methods. Horn and Schunck [2] pose flow as the minimisation of a global energy pairing

Three situations a fast platform meets, and what is actually computed in each



Fig. 1. Three situations a fast platform encounters, on held-out driving frames. Left: the frame with the region or regions a tracker would nominate. Right: the flow the decoder actually returns, computed only inside those regions. Top, a first encounter; middle, a turn in which the tracked region comes to overlap a second object; bottom, a new object appearing in a frame already being processed, which the decoder answers from cached state.

brightness constancy with a smoothness penalty on the field. Lucas and Kanade [3] solve brightness constancy in closed form over a window in which the flow is assumed constant. Both are local by construction, because the data term is a first-order expansion about zero displacement and holds only while motion is small relative to image structure. Brox et al. [4] add gradient constancy and a coarse-to-fine warping scheme, which extends the usable range but still loses a small object that moves far enough to vanish from the coarsest level. Large displacement of small structures is the failure every learned method since has been measured against.

Learned flow. FlowNet [5] replaced the energy with a network trained end to end on synthetic data, and showed that an explicit correlation layer between the two frames helps more than letting a generic encoder discover matching on its own. FlowNet 2.0 [6] closed the remaining gap to variational accuracy by stacking several such networks with warping between them, at the cost of a large model. PWC-Net [7] and LiteFlowNet [8] put the classical pyramid back inside the network: flow from the coarser level warps the second frame and a cost volume is built over a bounded displacement range around that warp, which is what made these models compact enough to be practical. The cascade’s own weakness is that an error made at a coarse level is never revisited. RAFT [9] removed the cascade by building one all-pairs correlation volume at $1/8$ resolution and reading from it repeatedly with a recurrent unit, refining a single estimate at fixed resolution; it set the accuracy standard the field still works

against, at considerable cost. GMFlow [10] instead treats correspondence as global matching, with cross-attention over the whole feature map producing flow directly and refinement reduced to a correction.

Real-time flow. NeuFlow [11] keeps global matching but runs it at $1/16$ behind a shallow backbone, then refines with cheap recurrent updates. NeuFlow v2 [1] restructures the backbone and refinement and ends with a learned convex upsampler that predicts blending weights over a 3×3 neighbourhood of coarse flow. That upsampler is the stage we replace, and everything above it we inherit frozen.

Implicit representations. NeRF [12] showed that a coordinate-conditioned MLP can hold a scene and be evaluated anywhere in it, and SIREN [13] showed that the activation function bounds how much high-frequency detail such a network can hold at all. LIIF [14] made the discrete-to-continuous step for images: a local latent code plus a relative coordinate, decoded by one shared MLP, so a single trained model can be sampled at any resolution. AnyFlow [15] brought arbitrary-scale decoding to flow by predicting convex weights over candidate displacements, which is the formulation our head adopts. It is built on RAFT and reports no inference timings, leaving open whether the approach survives a real-time budget. InfiniDepth [16] does the continuous version for depth and contributes the gated hierarchical fusion of multi-scale features that our decoder uses.

Uncertainty. Kendall and Gal [17] separate aleatoric from epistemic uncertainty and train a heteroscedastic likelihood whose predicted scale is supervised only by the

error it explains. Ilg et al. [18] applied this to flow directly, comparing multi-hypothesis networks against a single predictive distribution and showing that the estimate is usable rather than decorative. PDC-Net [19] predicts a mixture density per correspondence and uses it to decide which matches to trust, which is the use we make of our per-query scale. We differ in predicting the scale per query rather than per pixel, so it is available for exactly the correspondences the consumer selected, and we evaluate it by selective prediction rather than by correlation alone.

The gap. Every method above emits a dense field at one fixed resolution, chosen when the network is built. Those that are queryable are queryable in scale, not in coordinate, and the implicit decoders that are queryable in coordinate have not been placed inside a network operating under a millisecond budget. Whether arbitrary-coordinate querying is affordable at speed is therefore open. That is what this paper measures.

III. Method

A. Inherited Pipeline

NeuFlow v2 extracts features at 1/8 and 1/16 resolution with a shallow CNN, applies cross-attention and global matching at 1/16 to establish an initial flow, refines it recurrently (one iteration at 1/16, eight at 1/8), and upsamples the resulting 1/8-resolution flow with a learned convex combination over a 3×3 neighbourhood. That upsampling stage, comprising the full-resolution stem that feeds it and the convolutions that predict a separate nine-way softmax for each of the 64 sub-pixels in every 8×8 cell, is 1,947,968 parameters.

We retain everything through the coarse flow, with all learned weights and all normalisation statistics frozen at their published values. We verify this per checkpoint: all 137 tensors shared with the baseline are bit-identical, so the comparisons that follow isolate the decoder.

B. Implicit Decoder

The decoder operates in two phases, which is what makes repeat queries cheap.

Phase 1, once per frame pair. The frozen pipeline produces the coarse flow and feature maps, which are cached as state.

Phase 2, once per query batch. For a query coordinate (x, y) the decoder bilinearly samples 3×3 windows from four sources: 1/8 context, 1/8 matching features, 1/16 features, and frame-1 features warped by the coarse flow. These are fused by a gated hierarchical MLP following InfiniDepth, and the result conditions a head predicting weights over ten candidates: the nine coarse-flow values in the local neighbourhood, plus a bilinear sample at the query position.

The output is a convex combination,

$$\mathbf{f}(x, y) = \sum_{i=1}^{10} w_i(x, y) \mathbf{c}_i(x, y), \quad \sum_i w_i = 1, \quad w_i \geq 0, \quad (1)$$

```
st = model.infer_coarse_state(img0, img1)

# arbitrary coordinates
flow = model.decode_queries(st, query_coords=q)

# or any output resolution
flow = model.decode_queries(st, tgt_h, tgt_w)

# with per-query confidence
flow, b = model.decode_queries(st, q,
                               return_uncertainty=True)
```

Fig. 2. The complete interface. The coarse state is computed once per frame pair; each further decode reuses it.

where $\mathbf{c}_i$ are the candidate displacements and w_i are softmax weights. Boundedness matters: the prediction cannot leave the convex hull of locally supported motion, so the decoder cannot hallucinate displacement its inputs do not support.

We initialise the head so that the softmax concentrates on the bilinear candidate. An untrained decoder is therefore equivalent to bilinear upsampling (0.011 px maximum deviation), and training can only depart from that known-good operating point by learning.

The decoder is 745,226 trainable parameters, 38% of the 1,947,968-parameter stage it replaces. The complete model is 7,826,794 parameters against the baseline’s 9,029,536, 13.3% smaller.

C. Uncertainty Head

An optional eleventh output predicts a per-query error scale b , trained under a Laplace likelihood jointly with the flow loss,

$$\mathcal{L}_u = \frac{|\mathbf{f} - \mathbf{f}^*|}{b} + 2 \log b. \quad (2)$$

Predicting a large b is therefore only worth it where the error really is large. The head is 65 parameters, zero-initialised so that $b = 1$ px before training. It costs no measurable inference time and attaches a confidence value to every returned correspondence.

D. Interface

The interface is two calls, Fig. 2: one coarse pass per frame pair, then any number of decodes against the cached state. Coordinates are continuous, so (312.7, 188.2) is as valid an argument as (312, 188).

IV. Experimental Setup

Evaluation. Virtual KITTI 2 [20] Scenes 18 and 20: 1,174 frame pairs, 460,573,660 valid pixels, per-pixel metrics. Both scenes are excluded from every training set, enforced in the data loader and verified at pair and frame level before each run. Cross-domain evaluation uses TartanAir [21] aerial sequences, which no model in this study was trained on.

Training. All runs are generated from a single template so that any two differ in exactly one variable: seed 1234, batch size 16, learning rate 2×10^{-4} under a OneCycle

300 sparse queries at detected corners — decoded in one 1.6 ms call (0.05% of a dense field)



Fig. 3. 300 queries at detected corners, decoded in one call: 0.05% of the points a dense field would compute. Producing the same 300 answers from the baseline means computing 479,232 and discarding 478,932 of them.

TABLE I
Latency per frame pair

Mode	Latency	Note
v2, full frame	33.3 ms	its only mode
v3, first sparse query	34.1 ms	level with v2
v3, repeat query	1.25 ms	27× cheaper
v3, dense output	37.1 ms	not the intended mode

schedule, loss weighting $\gamma = 0.8$, 4,096 supervision queries per image, refinement schedule matching the evaluation configuration, 100,000 steps, backbone frozen throughout.

Hardware. NVIDIA RTX 4060 laptop GPU, fp16, 384×1248 input unless stated. Where a figure is device-dependent we report each device separately.

V. Results

The three subsections that follow address the three objectives in order, after which we account for what they cost and characterise region-directed inference.

A. Objective 1: Queryable Output

The decoder takes a batch of continuous coordinates and returns flow at those coordinates and nowhere else (Fig. 3). A corner detector selects its own points, they enter one decode call, and cost scales with how many were asked for.

Sparse and dense are not two modes but one function evaluated at different point sets: sparse output reproduces the dense field to 0.00057px maximum deviation at matching coordinates, which we verify before every run. Decode latency is flat in the query count over the operating range, 1.254ms at $N=800$ and 1.285ms at $N=2,048$, being bounded by kernel launch overhead rather than arithmetic, so 2,048 points cost what 800 do.

B. Objective 2: Cheap Repeatability

The sparse path comprises a 32.8ms coarse pass, inherited unchanged, and a 1.25ms decode. Because global matching requires whole-image context, the coarse pass cannot be avoided; it accounts for 96% of a first query, which is why a first query costs what a baseline frame costs. We do not claim an advantage there.

The decisive property is state reuse. The baseline retains nothing between calls, so a second question about the

TABLE II
Error of the accepted set at each coverage level

Coverage	NeuFlow v3	NeuFlow v2
20%	0.480	2.324
40%	0.688	2.324
60%	0.798	2.324
80%	1.058	2.324
100%	2.266	2.324

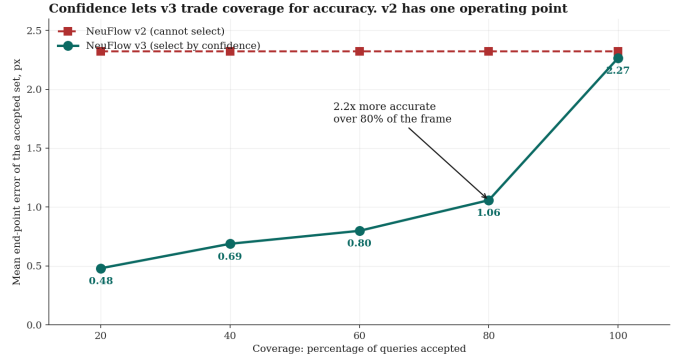


Fig. 4. Risk-coverage behaviour. The baseline has one operating point because it cannot indicate which outputs to trust; per-query confidence gives the decoder a curve.

same frame pair requires recomputing everything: 33.3 ms against 1.25 ms, a factor of 27. Three questions about one frame pair therefore cost the baseline 99.9 ms. The difference is structural rather than an optimisation gap, and it is what the two-phase factorisation exists to produce. In dense output mode, where nothing is reused, the baseline is the faster of the two at 33.3 ms against 37.1 ms.

C. Objective 3: Calibrated Confidence

Over 2,348,000 queries, actual error rises monotonically with predicted scale across all five equal-count bins, from 0.480px in the most confident to 7.100px in the least, a factor of 15, with Pearson correlation 0.318. The correlation is informative rather than tight, and we do not claim otherwise; what a consumer uses is the ordering, and the ordering is correct.

Mean error over every pixel is a single operating point, and it is the only one the baseline has, because it cannot indicate which of its outputs to trust. Given a per-query confidence, v3 has a curve (Table II, Fig. 4). Discarding the least confident 20% of queries reduces error on the remainder to 1.058px against the baseline’s 2.324px, more than twice as accurate over 80% of the frame. Retaining only the most confident fifth gives 0.480px.

The obvious objection is that the accepted 80% may simply be the easier 80%. We therefore scored both models on an identical query set, the matched subset the decoder accepts, and the result holds. Beyond that comparison the point is a capability rather than a margin: the baseline

TABLE III
Accuracy on held-out driving sequences

Model	Training data	EPE	1px	3px
NeuFlow v2	FlyingThings	2.324	77.6	89.8
NeuFlow v3	FlyingChairs	2.500	72.8	87.9
NeuFlow v3	+ VKITTI2	2.398	75.7	88.9
NeuFlow v3	+ Sintel	2.392	75.8	89.0
NeuFlow v3	+ uncertainty	2.384	76.1	89.0

TABLE IV
Processing a region instead of the frame, RTX 4060 laptop

Region	Latency	Speedup	EPE
Full frame	33.3 ms	1.0×	0.657
ROI, no margin	7.8 ms	4.3×	1.089
ROI +32 px margin	7.6 ms	4.4×	0.691
ROI +64 px margin	8.1 ms	4.1×	0.667
ROI +128 px margin	9.9 ms	3.4×	0.655

has no mechanism for selecting among its own outputs, so it cannot trade coverage for accuracy at any price.

D. The Cost in Accuracy

Table III reports what the three objectives cost. Every v3 configuration is less accurate than the baseline. The best is 2.6% worse on mean end-point error and 1.5 points worse on 1-pixel accuracy; the like-for-like comparison, trained on FlyingChairs alone as the baseline was trained on FlyingThings alone, is 7.6% behind. The cost is paid once, by a decoder that is 38% of the size of the stage it replaces, in a model 13.3% smaller overall.

The ordering across training mixes is monotonic, though the last three configurations differ by less than 0.02 px and we do not draw conclusions from their ranking under a single seed.

E. Cross-Domain Comparison

Region-directed inference is not specific to driving footage. Fig. 5 shows the same three situations in both domains under identical region geometry, which differ by roughly a factor of three in mean inter-frame motion.

On aerial sequences the decoder is ahead of the baseline at all seven margin settings tested, by 0.004 to 0.029 px (0.777 against 0.781 at full frame). We note this is the only unconfounded direct comparison in the study: every driving result is measured on a domain the decoder trained on and the baseline did not, whereas neither model has seen aerial imagery. The effect is small and measured on 40 pairs under one seed; it is consistent across seven settings, but the decoder remains slightly worse on large-motion failures (7.2% against 6.5%).

F. Region-Directed Inference

A platform that cannot afford full-frame flow must process only the region that matters. Table IV quantifies this for a 192×192 region of interest. Retaining full

TABLE V
The region saving is device-dependent; the accuracy cost is not. Same script, same 40 pairs, 192×192 region with a 32 px margin.

Device	Full frame	Region	Speedup	EPE cost
RTX 4060 laptop	33.3 ms	7.6 ms	4.4×	+0.034
Tesla V100	17.8 ms	15.1 ms	1.2×	+0.034

TABLE VI
EPE penalty in pixels against the full-frame result as a function of context margin, measured with the baseline network. Mean inter-frame motion is 26.6 px for driving and 9.26 px for aerial.

Driving (VKITTI2)		Aerial (TartanAir)	
Margin	Penalty	Margin	Penalty
0	+0.432	0	+0.424
16	+0.328	8	+0.120
32	+0.034	16	+0.091
64	+0.009	24	+0.098
128	−0.002	32	+0.053
		64	+0.012

resolution and processing roughly 14% of the frame area costs 0.034 px and runs 4.4× faster on the laptop GPU.

The saving, but not the cost, depends on the device (Table V). The same configuration is 4.4× faster on an RTX 4060 laptop and only 1.2× faster on a Tesla V100, whose kernel launch overhead dominates at this region size, so removing 86% of the pixels removes almost no work. The accuracy penalty is +0.034 px on both devices to three decimals, since it is the same computation. We report the two separately rather than averaging them: the technique pays on constrained hardware, which is the deployment target, and not on a datacenter GPU.

The context margin is not optional. Without one, error rises by 0.432 px and 24% of large-motion pixels fail outright, because global matching loses the context it uses to locate distant correspondences. We emphasise that this technique applies to any flow network including the baseline; it is a property of the architecture family, not of the queryable decoder.

G. The Margin Rule

If the margin requirement is set by displacement rather than by image size, it should scale with motion. We test this on aerial sequences whose mean displacement is a third of the driving case. Both domains begin from essentially the same penalty with no margin, 0.432 and 0.424 px, and both shed most of it once the margin reaches approximately one frame of motion (Table VI, Fig. 6), giving

$$\text{margin} \approx \text{inter-frame displacement} \approx \frac{\text{platform speed}}{\text{frame rate}}. \quad (3)$$

This can be evaluated before any flow is computed. The agreement is close up to that point and the residuals

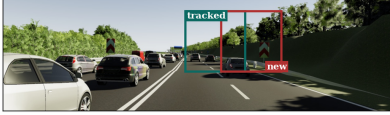
Driving

Virtual KITTI 2, 26.6 px mean motion

S1 First encounter — something worth flowing enters the field of view



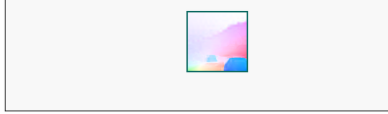
S2 Turn — the tracked region now overlaps a second object



S3 New object — a second region appears in a frame already being processed



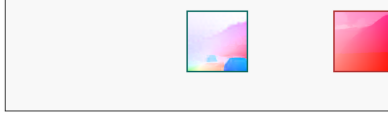
flow returned — computed only inside the regions



flow returned — computed only inside the regions



flow returned — computed only inside the regions



Aerial

TartanAir, 9.26 px mean motion

S1 First encounter something worth flowing enters the field of view



S2 Turn the tracked region now overlaps a second object



S3 New object a second region appears in a frame already being processed



flow returned — computed only inside the regions



flow returned — computed only inside the regions



flow returned — computed only inside the regions



Fig. 5. The same three situations in both evaluation domains, with identical region geometry and decoder configuration; only the domain changes. Left, driving (Virtual KITTI 2, 26.6 px mean inter-frame motion); right, aerial (TartanAir, 9.26 px). Teal marks the tracked region and red a region that appears later. Everything outside the marked regions is never computed. Neither model in this study was trained on aerial imagery.

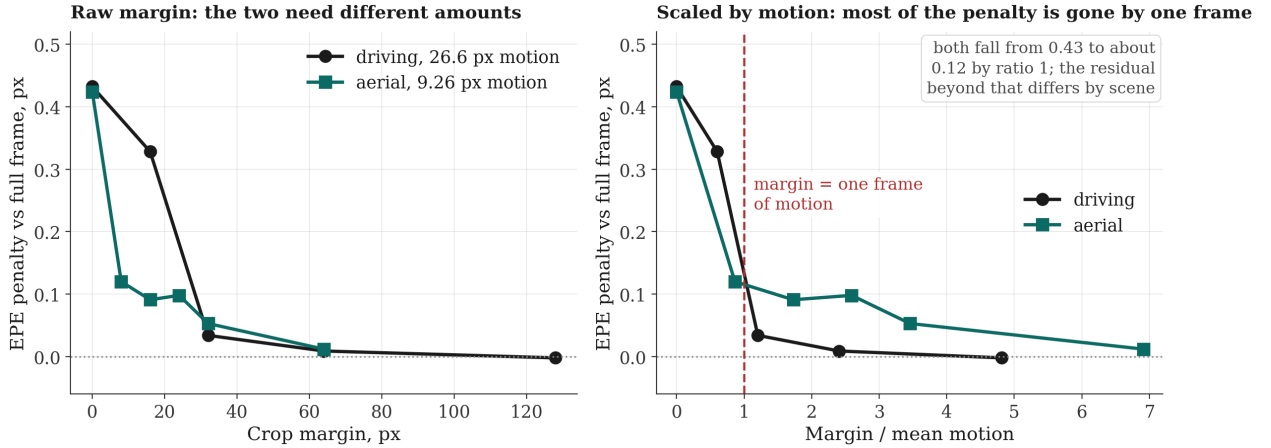


Fig. 6. Left: in raw pixels the two domains need different margins. Right: expressed as a multiple of mean inter-frame motion, both shed the bulk of the penalty by a ratio of one. Measured with the baseline network, since the effect belongs to the architecture family rather than to the decoder.

TABLE VII

Cost of a new region appearing mid-frame

Policy	Cost	EPE
v3, sparse queries	1.68 ms	2.10
v3, dense region	4.36 ms	1.77
v2, new cropped pass	7.29 ms	3.60
v3, new cropped pass	8.23 ms	3.61

beyond it differ by scene, so the relation predicts the scale of the required margin rather than the exact curve.

H. Answering an Unanticipated Query

A platform frequently discovers a second region of interest after a frame is already in flight. This is the case cheap repeatability exists for. Because the coarse state is cached, the decoder answers for 1.68 ms. A cropped

pipeline must execute an entire new pass, and is markedly less accurate doing so because a fresh crop has again lost global context (Table VII). Two disjoint crops cost more than one full frame, giving a simple policy: crop once, or not at all.

VI. Discussion

The contribution is an operating point rather than a leaderboard position. The decoder trades 2.6% of mean accuracy on its training domain for three properties the fixed-resolution design cannot express, from a model 13.3% smaller whose output stage is 38% of the size of the one it replaces.

For an application that consumes one complete dense field per frame and nothing else, the baseline remains the correct choice: it is more accurate and 11% faster in that mode, 33.3 ms against 37.1 ms. The decoder earns

its place when the consumer selects its own query points, revisits a frame, requires positions between pixels, or needs a confidence value with each correspondence.

We note that the third objective reframes the first two. Mean error over all pixels is the metric on which the decoder loses, and it is the only operating point available to a model without confidence. Permitted to abstain, the same decoder is more than twice as accurate as the baseline over most of the frame. The architecture costs 0.060 px on the mean; the confidence it makes possible returns 1.27 px over 80% of the frame.

VII. Limitations

Accuracy. The decoder is less accurate than the upsampler it replaces. The cause is structural: its finest input is at 1/8 resolution, so the evidence available to it varies little within an 8×8 cell, whereas the baseline’s upsampler reads the full-resolution frame directly. Three independent experiments converge on this diagnosis. Fourier positional encoding of the sub-cell offset produced no change, ruling out missing positional signal; querying below the input sampling rate matched bilinear upsampling exactly; and sub-pixel querying recovers little beyond interpolation. One mechanism accounts for all three.

Sub-pixel querying. For the same reason, evaluation between input pixels is presently closer to interpolation than to inference. The interface is exact and the mechanism is real, but the additional information recoverable between pixels is small while the decoder’s finest input remains at 1/8 resolution.

Embedded validation. All latency figures are from a laptop GPU. No measurement on embedded hardware has been performed, so the edge-deployment premise remains a design target rather than a result. The region speedup is additionally device-dependent, 4.4× on the RTX 4060 laptop against 1.2× on a Tesla V100, as reported above.

One primary evaluation domain. Virtual KITTI 2 is synthetic driving footage, partly answered by 40 aerial pairs from a second domain.

Statistical resolution. One seed per configuration, with checkpoint-to-checkpoint variation of up to 0.038 px. Differences below approximately 0.05 px are not resolved by this study.

VIII. Conclusion

We replaced the fixed upsampler of a real-time optical flow network with an implicit decoder answering queries at arbitrary continuous coordinates, and measured what the substitution costs and provides against three objectives. Queryable output is affordable: arbitrary continuous coordinates in $O(N)$ rather than $O(H \times W)$, for 2.6% of mean accuracy, from a model 13.3% smaller whose decoder is 38% of the size of the stage it replaced. Cheap repeatability is the structural win: a frame pair is processed once and cached, and every query after the first costs 1.25 ms against 33.3 ms, because the baseline keeps

no state between calls. Calibrated confidence is worth more than the mean it costs: the architecture costs 0.060 px on the mean, and abstaining on the least confident fifth returns 1.27 px over the remaining 80% of the frame.

We additionally characterised region-directed inference for fast platforms, establishing that a cropped region requires a context margin proportional to inter-frame displacement, a relation that holds across a threefold difference in motion scale and can be evaluated from platform speed and frame rate in advance.

Future work is directed by what this study could not settle. Embedded measurement on Jetson hardware is required before the deployment premise can be claimed, and nothing else here turns on a laptop GPU the way that claim does. A second seed per configuration would let the training mixes be ranked honestly, since three of the four differ by less than the noise floor. Evaluation above the input sampling rate, against ground truth at twice the input resolution, would test the arbitrary-resolution capability directly, and it is the one test the baseline structurally cannot take. Tiled refinement is the largest remaining speed opportunity, since refinement is 60% of runtime and all its operations are local, though it would benefit the baseline equally. Finally, registration on field and survey imagery would test the intended application and address the single-domain limitation directly.

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